FashionStylist: An Expert Knowledge-enhanced Multimodal Dataset for Fashion Understanding

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A Benchmark Baselines

A.1 Baselines for Task #1

- **FLUX.1 Kontext** [5]: A unified flow-matching model for image generation and editing that supports local edits and in-context transformations from text and image inputs.
- **Qwen-ImageEdit-2511** [11]: A large-scale image editing foundation model built upon Qwen-Image for precise instruction-following edits with strong text rendering and consistency preservation.
- **LongCat-ImageEdit-Turbo** [10]: A distilled image editing model that achieves high-quality editing with only 8 inference steps, enabling extremely low-latency generation.
- **Nano Banana2**[1]: A fast image generation and editing model from Google that combines advanced reasoning, subject consistency, and precise instruction following at Flash speed.

A.2 Baselines for Task #2

- **POG** [3]: A Transformer-based encoder-decoder model to generate professionalized outfits. FOM serves as the compatibility modeling backbone of POG, and POG further extends the pre-trained FOM by incorporating user preference signals for personalized outfit generation. We used the FOM model to complete the outfit.
- **CIRP** [6]: A representative multimodal pre-training model for bundle construction, where contrastive learning is incorporated for learning cross-item relationship.
- **CLHE** [7]: A representative multimodal model for bundle construction, where multimodal information and contrastive learning are introduced to alleviate the data insufficiency and cold-start problems.
- **DiFashion** [12]: A generative recommendation model based on SD-v2, which use classifier-free guidance to align the control conditions of the generated fashion images.

A.3 Baselines for Task #3

- **Gemma3-4B** [8]: A 4B-parameter open-weight multimodal model developed by Google, supporting vision-language understanding with a 128K context window.
- **Qwen2.5-VL-7B** [2]: A 7B-parameter vision-language model from the Qwen family, featuring dynamic resolution processing and enhanced visual grounding capabilities.
- **Qwen3-VL-4/8B** [1]: A series of vision-language models from the Qwen family, available in 4B and 8B dense variants, featuring enhanced interleaved multimodal reasoning with a native 256K-token context window.
- **Gemini3.1-Pro-preview** [9]: A family of highly capable multimodal models developed by Google DeepMind, representing state-of-the-art proprietary performance across vision-language tasks.

B Implementation Details

B.1 Details for Task #1

We use a unified inference interface for all baselines. Specifically, FLUX.1-Kontext and LongCat-ImageEdit-Turbo are executed with the `diffusers` framework[2], while Qwen-ImageEdit-2511 is run with the DiffSynth-Studio backend[3] for compatibility with its native image-edit pipeline.

For FLUX.1-Kontext, we perform inference in bf16 with 28 denoising steps, a guidance scale of 3.5, a maximum text length of 512, and a fixed random seed of 42. During LoRA training, we set the rank to 16 and the LoRA alpha to 16, use batch size 1 with gradient accumulation steps 4, and optimize the trainable parameters using

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[1]https://ai.google.dev/gemini-api/docs/models/gemini-3.1-flash-image-preview

[2]https://huggingface.co/docs/diffusers/index
[3]https://github.com/modelscope/DiffSynth-Studio

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AdamW with 8-bit optimizer states, a learning rate of 1×10^{-4}, a constant learning-rate schedule, and 200 warm-up steps.

For LongCat-ImageEdit-Turbo, we use bf16 inference with 8 denoising steps, a guidance scale of 1.0, and seed 42, inference uses 50 denoising steps and a guidance scale of 4.5. For LoRA training, we use a learning rate of 1×10^{-4}, 5 epochs, dataset repeat 50, gradient accumulation steps 1, and a maximum pixel budget of 1024×1024. We set the LoRA rank to 16 and apply LoRA to the default target modules in the DiT backbone. Negative prompting is enabled during inference.

For Qwen-ImageEdit-2511, we use bf16 precision, 28 denoising steps, a true CFG scale of 4.0, and seed 42. For its LoRA variant, we use the same training setup as LongCat, including a learning rate of 1×10^{-4}, 5 epochs, dataset repeat 50, gradient accumulation steps 1, and a maximum pixel budget of 1024×1024. We set the LoRA rank to 16 and apply LoRA to the default target modules in the DiT backbone. Negative prompting is also enabled during inference.

B.2 Details for Task #2

Below, we summarize the implementation details of the baselines used for Task #2.

For **POG**, we use frozen FashionCLIP image and text embeddings as item representations and fuse the two modalities with an MLP. A masked outfit is then encoded by a Transformer encoder to capture item compatibility within the outfit. The model is trained with a contrastive masked-item prediction objective using sampled negatives, and optimized with AdamW and cosine annealing for 30 epochs. The best-performing setup uses lr = 2×10^{-4}, weight decay = 1×10^{-5}, dropout = 0.15, embed_dim = 256, hidden_dim = 256, 4 Transformer layers, 8 attention heads, ff_dim = 256, 6 negative samples, `concat` fusion, and batch size = 32.

For **CIRP**, we first fine-tune a FashionCLIP-based multimodal encoder on FashionStylist image-text pairs using the image-text contrastive (ITC) loss, for 10 epochs with a batch size of 64, a learning rate of 3×10^{-5}, and a weight decay of 0.05. We then train LightGCN on the full item-transition graph with a learning rate of 5×10^{-4}, an embedding size of 64, two graph layers, ℓ_2 regularization of 1×10^{-6}, 400 epochs, and a batch size of 64. The learned graph is used for relation pruning with a drop rate of 0.1. Next, we initialize the CIRP encoder with the first-stage encoder and further train it on the pruned graph using both the ITC loss and the cross-item contrastive (CIC) loss. In this stage, we use a batch size of 64, an initial learning rate of 1×10^{-5}, 3000 warmup steps, a weight decay of 0.05, and 31 epochs. Finally, the learned item representations are evaluated with ItemKNN on the bundle completion task.

For **CLHE**, we adopt Factor-wise Dropout and Item Dropout as the item-level and bundle-level augmentation strategies, respectively. During training, 50% of the items in each bundle are retained. For the loss function, we set $\alpha_1 = 0.1$, $\alpha_2 = 1.0$, and $\beta = 1 \times 10^{-5}$. The temperature parameters for the item-level and bundle-level contrastive losses are set to 0.2 and 0.1, respectively. The learning rate is 1×10^{-3}, the batch size is 128, and the model is trained for 50 epochs.

For **DiFashion**, we follow the original training and inference settings in [12]. To adapt the model to Task #2, we use ResNet50 to encode item images, and then perform retrieval based on cosine similarity to compute Recall and NDCG.

B.3 Details for Task #3

For the Instruct and Think settings, we implement them via vLLM [4]. The batch size is set to 4, the temperature is set to 0.3, the top-k value is set to 50, and the maximum number of generated tokens is set to 1024, and the context length is set to 16384. For Qwen-VL models with visible reasoning, we enlarge the generation budget to 12288 tokens and the context length to 32768 to accommodate longer reasoning traces.

For the supervised fine-tuning setting (*i.e.,* Sft), we implement it via Unsloth [5] and use the same configuration for all models. Specifically, the number of training epochs is set to 2, the batch size is set to 1, and the number of gradient accumulation steps is set to 4. The learning rate is set to 2×10^{-4}, the warmup ratio is set to 0.03, the weight decay is set to 0.01, and the maximum sequence length is set to 8192. For LoRA [4], we set the rank to 16, the scaling factor to 16, and the dropout rate to 0.0. The checkpoints are saved every 100 steps. During inference after fine-tuning, the generation temperature is set to 0.3. The best checkpoint for each model is selected on the validation split using the modified-item index accuracy on modified samples as the model selection criterion.

C Prompt Template

Only Task #2 and Task #3 in this paper involve textual prompting. For Task #2, we adopt two prompt templates for different model groups: a CLIP-based prompt and a unified prompt for Qwen, ImageCat, and Nano Banana2, as illustrated in Box 1 and Box 2, respectively. The textual prompt for Task #3 is presented in Box 3.

[4] https://vllm.ai
[5] https://unsloth.ai

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Box 1: Prompt Template for Task #2 (FLUX.1 Kontext)

Prompt:

```
Extract only the {category} as a flat lay on pure white.
{category_hint}
Keep exact color, fabric, shape, and details. One item only.
```

Category Hints: *Outerwear*: Show full collar and sleeves. *Inner Top*: Show full torso and neckline. *Mid Layer Top*: Show full torso, neckline, and sleeves. *Bottom*: Show full waistband to hems. *One Piece*: Show full garment from neckline to hem. *Shoes*: Show both shoes as one matching pair. *Bag*: Keep the full bag and straps. *Accessory*: Keep the full accessory; keep pairs together.

Box 2: Prompt Template for Task #2 (Qwen-ImageEdit / LongCat-Turbo)

Prompt:

```
A professional e-commerce product flat-lay photography of the isolated {category}
worn by the model in the source image. Top-down view, completely cut out and
perfectly positioned on a pure solid white (#FFFFFF) background.
{shape_constraint} {quantity_constraint}
Maintain the exact original fabric, texture, color, and structural details from
the source. High-end clean studio lighting, pristine and highly detailed, sharp
edges without any bleeding.
```

Negative Prompt:

```
(human feature, skin, face, hand, arm, leg, model body:1.5), original background,
environmental shadows, other garments, overlapping clothes, noise, artifacts,
duplicate items, multiple instances, hallucinated patterns, missing parts.
```

Shape Constraints: *Outerwear*: reconstruct the full shape including collar and sleeves. *Inner Top*: reconstruct the full torso and neckline without outerwear occlusion. *Mid Layer Top*: reconstruct the full torso, neckline, and sleeves without occlusion. *Bottom*: reconstruct the full shape from waistband to hems. *One Piece*: reconstruct the complete continuous garment from neckline to hem. *Shoes*: keep the visible shoe product complete and natural. *Bag*: retain the full structural volume and straps/handles. *Accessory*: preserve material details; keep paired items together when applicable.
Quantity Constraints: *Outerwear / Inner Top / Mid Layer Top / Bag*: present exactly one single un-duplicated item. *Bottom*: present exactly one single pair of bottoms, with no duplicated pants or skirts. *One Piece*: present exactly one complete continuous garment. *Shoes*: do not invent extra shoes or force a pair if the source image only supports a single shoe product. *Accessory*: present the accessory clearly; if it is a naturally paired item, keep the pair together.

Box 3: Prompt Template for Task #3

You are a professional fashion evaluator and stylist.

Input: You will receive exactly one JSON object + a list of images. `images`: placeholder array (the actual images are provided in order); `candidate_style / candidate_season / candidate_occasion`: complete lists of allowed labels. Any `mod_index` must refer to a 0-based index of the corresponding image.

Task: Analyze the entire outfit holistically (silhouette, color, layering, fabric, pattern, mood). Follow this exact internal order: (1) Draft a concise `outfit_summary`. (2) Select `outfit_style` that best matches the summary (exactly one from `candidate_style`). (3) Select `season` and `occasion` from their candidate lists. (4) Decide if exactly one clear outlier item breaks the overall logic; if yes, set `need_to_modify=1` and mark that single `mod_index`; otherwise set `need_to_modify=0` and `mod_index="NONE"`.

Decision Rules:

A. Overall Outfit Judgment — Evaluate holistically using silhouette, color, layering, material, pattern, accessories, and mood. Base the final judgment on the coherent majority of items.

B. Label Alignment — `outfit_summary`, `outfit_style`, `season`, and `occasion` must describe the same outfit. Choose labels best supported by visible evidence; do not invent labels outside the candidate sets.

C. Outlier Judgment — Use `need_to_modify=1` only when exactly one item is clearly mismatched. Use `need_to_modify=0` when evidence is weak, ambiguous, or multiple items would need changes.

Output: Return exactly one JSON object. `outfit_summary`: one sentence, 15–40 words, must not copy the style label.

JSON Schema:

```
{
 "outfit_summary": "string",
 "outfit_style": "from candidate_style",
 "season": "from candidate_season",
 "occasion": "from candidate_occasion",
 "need_to_modify": 0 or 1,
 "mod_index": "NONE" or integer
}
```

Example — no conflict:

```
{"outfit_summary": "A cropped denim jacket
 paired with straight-leg trousers,
 creating a relaxed yet neat everyday
 autumn-winter look.",
 "outfit_style": "Denim Casual",
 "season": "Autumn-Winter",
 "occasion": "Daily",
 "need_to_modify": 0,
 "mod_index": "NONE"}
```

Example — with conflict (index 2):

```
{"outfit_summary": "A long coat layered
 with knitwear and wide-leg trousers,
 giving a sharp and composed look for
 winter office wear.",
 "outfit_style": "Elevated Commuter",
 "season": "Winter",
 "occasion": "Business",
 "need_to_modify": 1,
 "mod_index": 2}
```

Now analyze the given sample. Return JSON only.